Choosing Linear, Quadratic, or Piecewise Models

Evidence that a situation is linear, evidence that it is quadratic, evidence that it is piecewise, and comparing two models against each other. Before you compute, decide which kind of model the evidence supports:

- **Linear** — equal steps in the input produce *equal* steps in the output (constant first differences), or the story names one fixed rate that never changes.
- **Quadratic** — the first differences change but change by the same amount each time (constant second differences), or the story involves a product of two quantities that both depend on the input (area, revenue, projectile height).
- **Piecewise** — the rule itself changes at a stated boundary: "first 2 hours… after that…". One formula cannot cover both sides.
- Say what your evidence is, then build the model, then answer.

1. A tutoring service posts this price table.

Hours h	0	1	2	3
Total cost (dollars)	25	40	55	70

The first differences are equal, so a linear model fits. Write the model, then use it to find the total cost of 6 hours of tutoring, in dollars.

Answer: ________________

2. Using the same tutoring model from problem 1, how many hours were bought if the total cost was 175 dollars?

Answer: ________________

3. A ball is launched straight up. Its height above the ground is recorded each second.

Time t **(s)**	0	1	2	3
Height (ft)	0	48	64	48

The second differences are constant, and the model that fits is $h(t) = -16t^2 + 64t$. Find the height of the ball at $t = 3.5$ seconds.

Answer: ________________ ft

4. Using the same height model $h(t) = -16t^2 + 64t$ from problem 3, find every time at which the ball is at ground level. Give both values of t, in seconds.

Answer: ________________

5. A parking garage charges 4 dollars for the first 2 hours and 3 dollars for each additional hour after that.

Find the cost, in dollars, of parking for 5 hours.

Answer: ________________

6. A pool is being filled. The depth is measured each minute.

Minutes x	1	2	3	4
Depth (cm)	7	12	17	22

(a) Decide which kind of model fits, write it, and predict the depth at $x = 10$ minutes.

(a) ____________

(b) Explain what feature of the table shows the relationship is linear rather than quadratic.

7. Return to the parking garage from problem 5: 4 dollars for the first 2 hours, then 3 dollars per additional hour.

A driver paid 22 dollars. How many hours did the car stay in the garage?

Answer: ________________

8. A shop finds that when it sets the price of a hoodie at p dollars, it sells $60 - 2p$ hoodies. Its revenue is therefore $R(p) = p(60 - 2p)$.

Find both prices at which the revenue is 400 dollars.

Answer: ________________

9. Two gyms both charge a fixed joining fee plus a fixed monthly rate — so each is a linear model.

Gym A: 20 dollars to join, then 8 dollars a month. Gym B: 44 dollars to join, then 5 dollars a month.

After how many months x do the two plans cost the same, and what is that cost y?

Answer: ________________

10. A quadratic model $y = x^2 - 4x + 7$ and a linear model $y = 2x + 2$ describe the same quantity in two different studies.

Find every point (x, y) where the two models agree.

Answer: ________________

11. A population is counted each year.

Year t	0	1	2	3
Population P	5	9	17	29

Rae looks at the first step, sees an increase of 4, treats the growth as a constant rate and predicts $P = 29$ at year 6.

Explain what Rae's evidence actually shows about the model, then use the correct model $P(t) = 2t^2 + 2t + 5$ to find the population at year 6.

Answer: ________________

12. Synthesis challenge. A rideshare charges a 3 dollar flag drop, then 2 dollars per mile for the first 6 miles, and 1.25 dollars per mile for every mile after that.

(a) Find the cost of a 10 mile ride, in dollars.

(a) ____________

(b) A ride costs 27.50 dollars. How many miles was it?

(b) ____________

(c) Explain why no single linear function can describe this fare for every distance, and describe what the graph of the fare looks like.